LAPORAN PRAKTIKUM

TEKNOLOGI IOT II

Komunikasi Asinkron dan WebSockets, Topologi Jaringan Terdesentralisasi (ESP Now dan Mesh)

Disusun Oleh:

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TRI – 3D

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TEKNOLOGI REKAYASA INTERNET

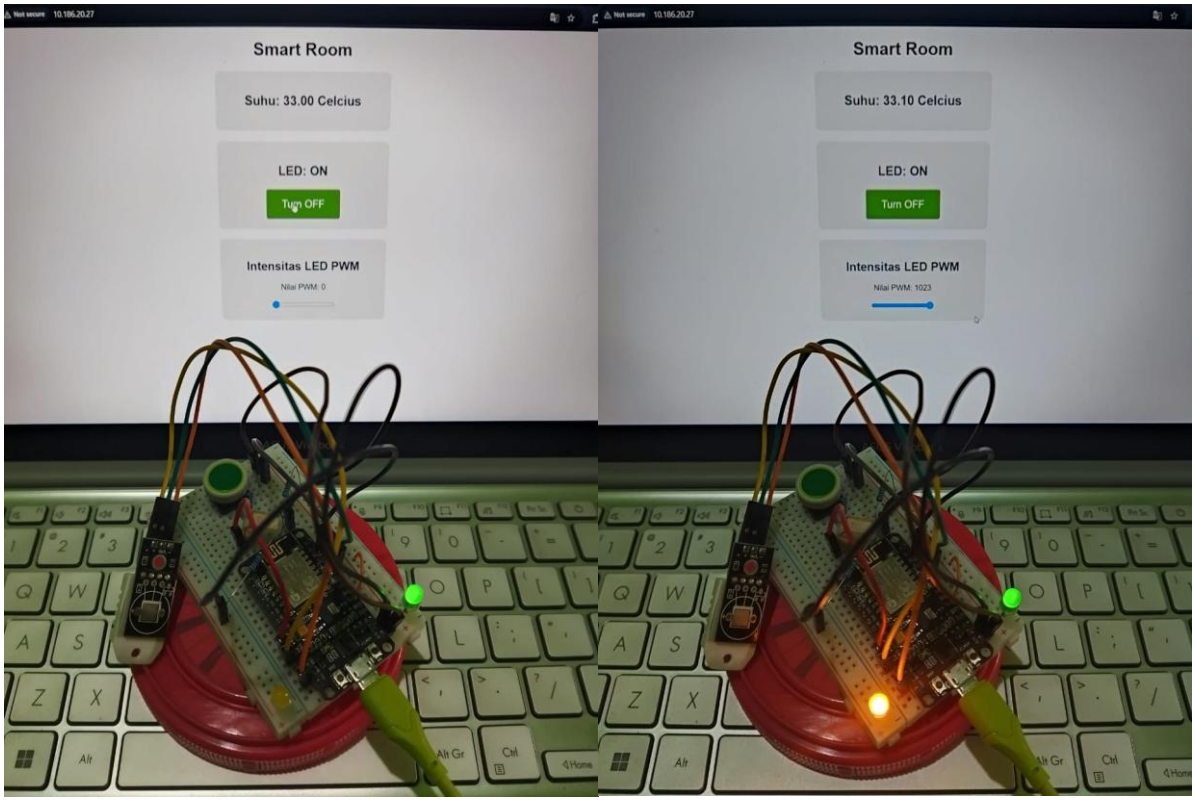
JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI LAMPUNG

2026/2027

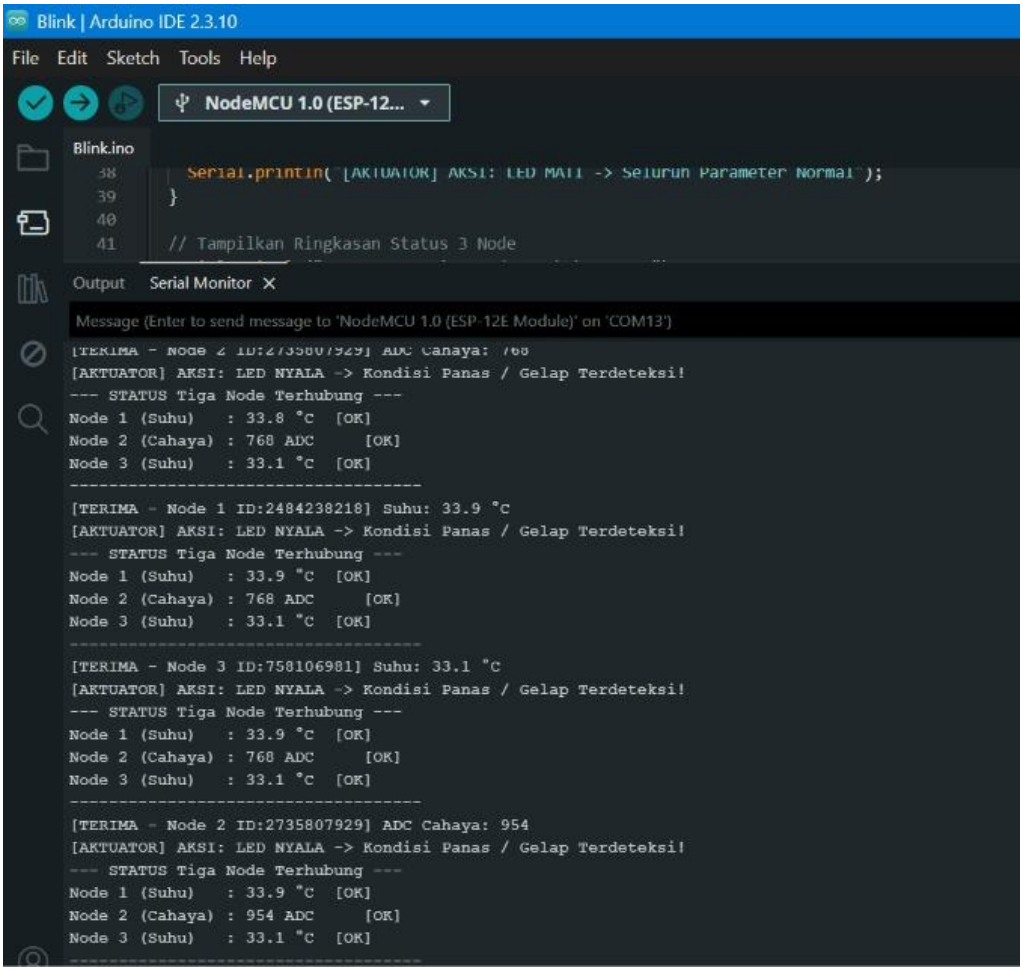
4.7 Tugas: Sistem Kendali Intensitas Cahaya (PWM Slider) via *WebSockets*

Pada Praktikum ini bertujuan untuk meningkatkan sistem kendali IoT dari saklar digital *ON/OFF* biasa menjadi sistem kendali analog (Dimmer) menggunakan modulasi PWM yang dioperasikan melalui elemen *slider* HTML dan WebSockets, di mana pergeseran tuas pada layar *smartphone* akan mengirimkan format teks string khusus ke NodeMCU untuk langsung meredupkan atau menerangkan cahaya fisik LED secara mulus dan *real-time* tanpa mengganggu fungsi sensor DHT maupun kontrol perangkat lainnya.

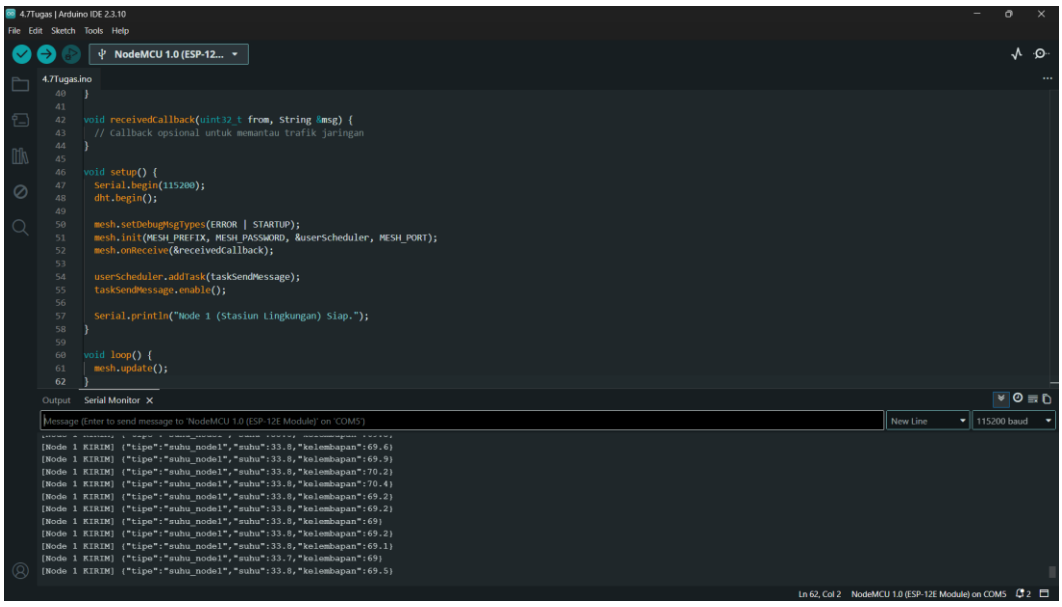


5.7 Tugas: Sistem Telemetri & Otomasi Terdistribusi Berbasis painlessMesh

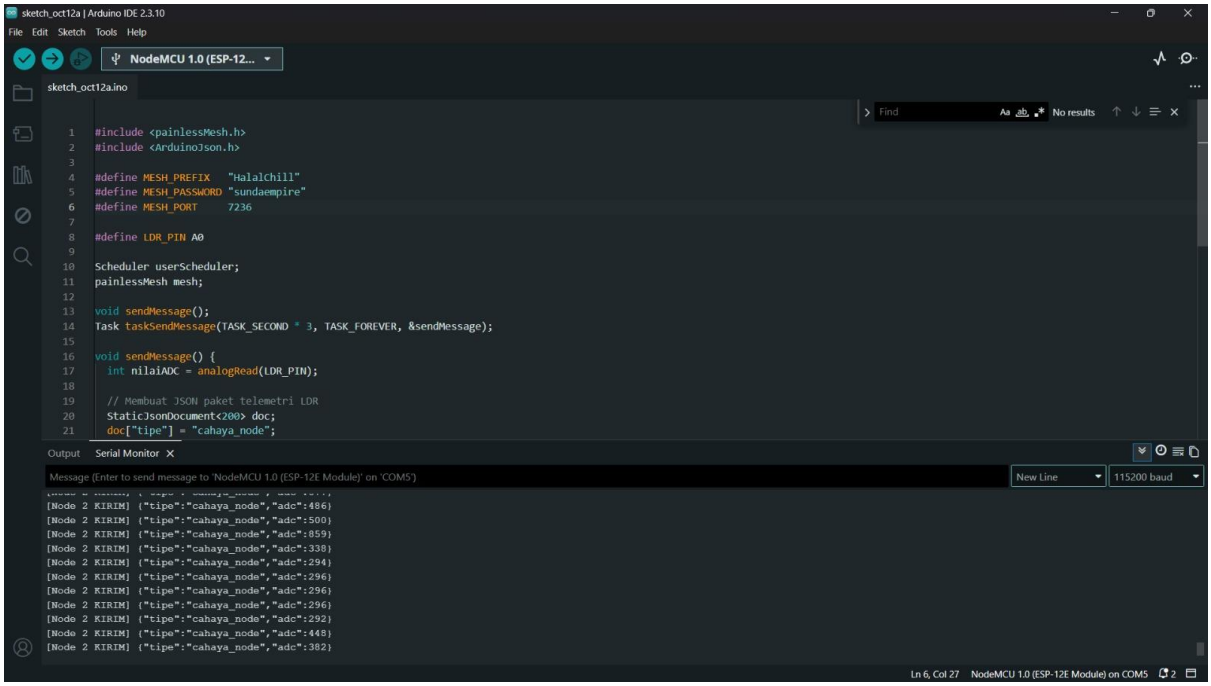
Aktuator :



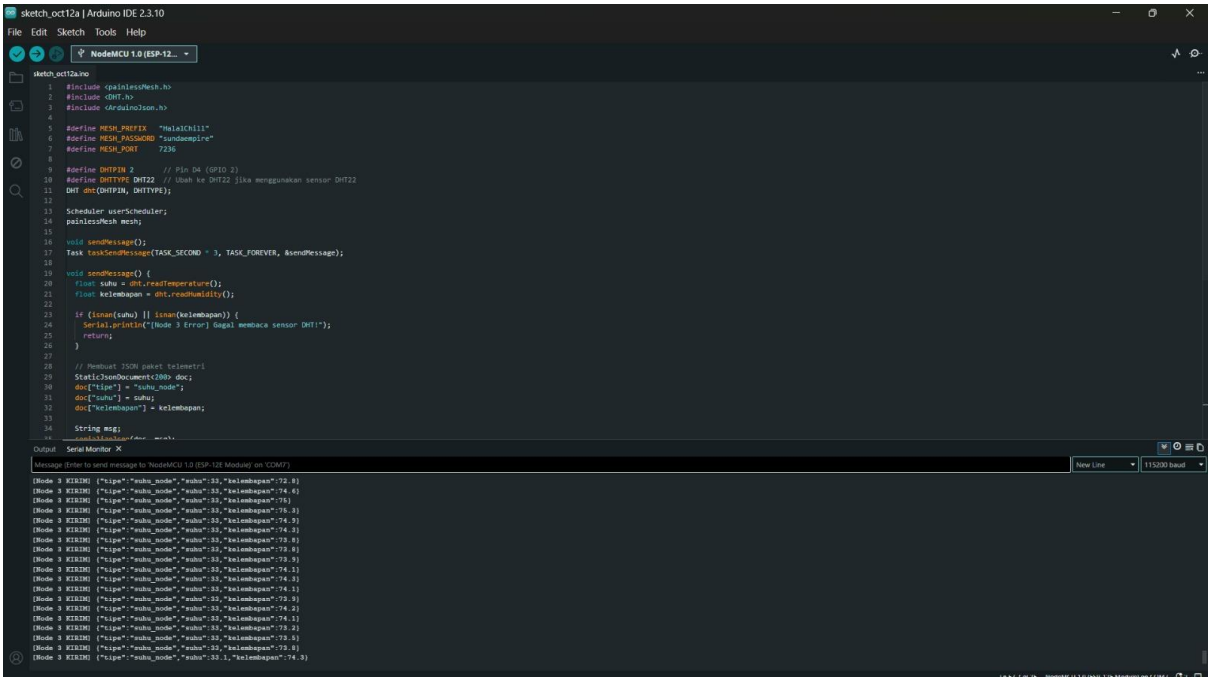
Node 1 :



Node 2 :


sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
#include <painlessMesh.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "HalalChill"
#define MESH_PASSWORD "sundaempire"
#define MESH_PORT 7236
#define LDR_PIN A0
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
void sendMessage() {
int nilaiADC = analogRead(LDR_PIN);
// Membuat JSON paket telemetri LDR
StaticJsonDocument<200> doc;
doc["tipe"] = "cahaya_node";
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
New Line
115200 baud
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":486}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":500}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":859}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":338}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":294}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":296}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":292}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":448}
[Node 2 KIRIM] {"tipe":"cahaya_node","adc":382}
Ln 6, Col 27 NodeMCU 1.0 (ESP-12E Module) on COM5

Node 3 :


sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
#include <painlessMesh.h>
#include <DHT.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "HalalChill"
#define MESH_PASSWORD "sundaempire"
#define MESH_PORT 7236
#define DHTPIN 2 // Pin D4 (GPIO 2)
#define DHTTYPE DHT22 // Ubah ke DHT22 jika menggunakan sensor DHT22
DHT dht(DHTPIN, DHTTYPE);
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
Task taskSendMessage(TASK_SECOND * 3, TASK_FOREVER, &sendMessage);
void sendMessage() {
float suhu = dht.readTemperature();
float kelembapan = dht.readHumidity();
if (isnan(suhu) || isnan(kelembapan)) {
Serial.println("[Node 3 Error] Gagal membaca sensor DHT!");
return;
}
// Membuat JSON paket telemetri
StaticJsonDocument<200> doc;
doc["tipe"] = "suhu_node";
doc["suhu"] = suhu;
doc["kelembapan"] = kelembapan;
String msg;
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
New Line
115200 baud
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":72.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.6}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":76}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":76.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.3}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.9}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.2}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":74.1}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.2}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.5}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33,"kelembapan":73.8}
[Node 3 KIRIM] {"tipe":"suhu_node","suhu":33.1,"kelembapan":74.3}